Lineshape Correlation Coefficient (LCC)

A 2D-HSQC similarity measure synthesized from the three methods already in this repository — the Bodis *et al.* bin method, the Castillo *et al.* quad-tree, and the Pierens *et al.* nearest-neighbour matcher — designed to keep the bin method's discrimination on dense protein `1H-15N` fingerprints while removing its shift-brittleness and without the tree/NN saturation. Implemented in `../hsqc_lcc.py`.

One line: render each spectrum to a common grid, blur it by the physical NMR linewidth, and take the mean-centred normalized cross-correlation (Pearson / ZNCC) of the two images at zero lag.

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1. Setup and preprocessing

Two spectra $a \in \{x, y\}$ are compared inside a common window $\Omega = [\delta_{\text{lo}}^{F2}, \delta_{\text{hi}}^{F2}] \times [\delta_{\text{lo}}^{F1}, \delta_{\text{hi}}^{F1}]$ (direct dimension $F2$, e.g. ^{1}H; indirect dimension $F1$, e.g. ^{15}N). Each processed point p has shifts $\delta_{F2}(p), \delta_{F1}(p)$ and intensity $I_a(p)$.

With the default `clip` baseline negative intensities are zeroed, and every point is weighted by its local integration area so spectra of different digital resolution integrate consistently:

$$w_a(p) \;=\; \max(I_a(p), 0) \;\cdot\; |\Delta\delta_{F1}(p)| \;\cdot\; |\Delta\delta_{F2}(p)|.$$

(This is exactly the area weighting of the bin method. The mass is normalized to $\sum_p w_a(p) = 1$; the final ZNCC is invariant to this scale, so normalization only guards against overflow.)

2. Rendering to a shared grid

The window is discretized once into a fixed grid shared by **both** spectra: K bins along $F1$ of width Δ_1 (default `step_f1` $= 0.10$ ppm) and L bins along $F2$ of width Δ_2 (default `step_f2` $= 0.01$ ppm). Each spectrum is rendered by summing its weighted points into cells (k, l):

$$R_a[k,l] \;=\; \sum_{p\,:\,\delta(p)\,\in\,\text{cell}(k,l)} w_a(p), \qquad k = 1, \dots, K, \;\; l = 1, \dots, L.$$

Rendering both spectra onto the *same* edges is what makes the two images directly comparable regardless of their native resolutions.

Optional intensity compression. To stop a single dominant amide from dominating the correlation, the rendered intensities may be raised to a power ρ (`--p`, default $\rho = 1$, i.e. off):

$$R_a[k,l] \;\leftarrow\; R_a[k,l]^{\rho}.$$

3. Lineshape blur

The discrete image is convolved with a separable Gaussian whose width is the physical NMR linewidth *plus* the expected chemical-shift drift, one σ per axis:

$$\sigma_{F2} = \sqrt{\ell_{F2}^2 + d_{F2}^2}, \qquad \sigma_{F1} = \sqrt{\ell_{F1}^2 + d_{F1}^2},$$

with linewidth ℓ and expected drift d (defaults $\sigma_{F2} = 0.03$, $\sigma_{F1} = 0.30$ ppm). The 1-D kernel, with $s = \sigma/\Delta$ the width in pixels and radius $r = \lceil 3s \rceil$, is

$$g_\sigma(m) \;=\; \frac{\exp(\,-\,m^2/2s^2)}{\sum_{m'=-r}^{r}\exp(\,-\,m'^2/2s^2)}, \qquad m=-r,\dots,r,$$

and the blurred ("lineshape-rendered") image is the separable convolution

$$G_a \;=\; g_{\sigma_{F1}} \;*_{F1}\; \big(g_{\sigma_{F2}} \;*_{F2}\; R_a\,\big).$$

The continuous Gaussian is precisely what replaces the bin method's *hard* bin edges with a *soft* response (see §6).

4. Mean-centring and the similarity score

Each image is centred by subtracting its global mean

$$\bar{G}_a = \frac{1}{KL}\sum_{k,l} G_a[k,l], \qquad \widehat{G}_a[k,l] = G_a[k,l] - \bar{G}_a,$$

and the similarity is the mean-centred normalized cross-correlation at **zero lag** (the Pearson correlation coefficient of the two flattened images), clamped to $[0,1]$:

$$\boxed{S(x,y) \;=\; \max\left(0,\; \frac{\sum_{k,l}\widehat{G}_x[k,l]\,\widehat{G}_y[k,l]}{\sqrt{\sum_{k,l}\widehat{G}_x[k,l]^2}\;\sqrt{\sum_{k,l}\widehat{G}_y[k,l]^2}}\right)}$$

No shift search / alignment is performed — this is deliberate (§6).

5. Properties

Self-similarity is exactly 1. For $y=x$ the numerator and each factor of the denominator are the same sum $\|\widehat{G}_x\|^2$, so

$$S(x,x) = \frac{\|\widehat{G}_x\|^2}{\sqrt{\|\widehat{G}_x\|^2}\,\sqrt{\|\widehat{G}_x\|^2}} = 1,$$

and the $\min(1,\cdot)$ clamp in the code absorbs the last-ULP floating-point wobble so an identical spectrum returns `1.0` bit-exactly.

Symmetry. $S(x,y) = S(y,x)$ (the score is an inner product).

Range. As a Pearson coefficient $S \in [-1,1]$ before clamping; the clamp maps anticorrelation to 0, giving $S \in [0,1]$.

Degenerate guard. A flat image ($\widehat{G}_a \equiv 0$) has zero denominator and returns 0; the `_window` step already excludes empty windows upstream.

6. Why it beats the existing methods

The three source methods fail in *opposite* ways: the bin method is shift-**brittle** (a peak straddling a bin edge splits its mass, so a small titration drift drops the score in a discontinuous jump), while the tree and NN methods are shift-**blind** (in a dense fingerprint every peak has a near neighbour, so both saturate near 1). LCC sits between them via three mechanisms:

(a) Graded shift tolerance. Model a peak as a point mass; after the blur it is a Gaussian of width σ. Two copies of one peak displaced by (Δ_H, Δ_N) contribute to the (centred, normalized) correlation a factor

$$\exp\left(-\frac{\Delta_H^2}{4\sigma_{F2}^2} - \frac{\Delta_N^2}{4\sigma_{F1}^2}\right),$$

which is **smooth and monotone** in the drift — no bin-edge discontinuity. A physical titration drift ($\Delta_H \sim 0.01$–0.05, $\Delta_N \sim 0.2$–0.8 ppm) is a fraction of σ and costs little; a random relocation ($\Delta \gg \sigma$) drives the factor toward 0. The single knob σ sets the tolerance scale.

(b) Coverage penalty from mean-centring. Writing the score as a covariance,

$$S \propto \mathrm{cov}(G_x, G_y) = \frac{1}{KL}\sum_{k,l}(G_x[k,l] - \bar{G_x})(G_y[k,l] - \bar{G_y}),$$

a cell where one spectrum has a peak (value above its mean) and the other is empty (value below its mean) contributes a **negative** product and *lowers* S. So intensity that is not co-located is actively penalised — a differently scattered protein cannot hide behind coincidental proximity the way it does under NN's one-way nearest-neighbour distance. This mean-centring step is worth $\approx +0.17$ of separation over an un-centred overlap.

(c) Zero lag on purpose. Adding a global shift search (FFT/block matching) would let a different protein slide into registration and re-saturate — the same failure as tree/NN. Scoring at zero lag keeps the position information that carries the discrimination.

7. Parameters

symbol	flag	default	meaning
σ_{F2}	`--sigma-f2`	0.03 ppm	^{1}H linewidth + drift blur width (main $F2$ tolerance)
σ_{F1}	`--sigma-f1`	0.30 ppm	^{15}N/^{13}C linewidth + drift (main tuning lever)
Δ_2	`--step-f2`	0.01 ppm	$F2$ render pixel width (≥ 3 px per σ; score grid-invariant)
Δ_1	`--step-f1`	0.10 ppm	$F1$ render pixel width
ρ	`--p`	1.0	intensity compression exponent (off)

Fixed by design: `clip` negatives, centring **on**, shift search **off**.

8. Benchmark

Reference PRL3 experiment 2 (`1H-15N` HSQC) versus the same protein plus ligand (experiments 4–14, should score high) and a different protein (OAA 103, should score low). Window $F2$ 6.5–10, $F1$ 105–130 ppm. Separation $\Sigma = \overline{S}_{\text{same}} - S_{\text{diff}}$; margin $= \min_e S_{\text{same}} - S_{\text{diff}}$.

method	self	mean same	different	Σ	margin
Bin (Bodis 2009)	1.00	0.79	0.49	0.29	0.24
Bin + 45°	1.00	0.82	0.57	0.25	0.20
Quad-tree (Castillo 2013)	1.00	0.90	0.87	0.03	−0.01
Nearest-neighbour (Pierens 2012)	1.00	0.99	0.96	0.04	0.03
LCC (this work)	1.00	**0.94**	**0.18**	**0.75**	**0.71**

LCC separates same-protein from different-protein spectra $\approx 2.6\times$ better than the previous best, pushing the different protein below *every* same-protein score. The gain holds across the whole physical blur range ($\Sigma = 0.71$–0.77 at $\sigma = 0.02$–0.04 / 0.20–0.40 ppm; still 0.36 when coarsened to the bin method's own resolution 0.10 / 1.0), so it is not an artefact of one tuned parameter. Numbers and plot in `../results/`.

9. Relationship to the source methods

ingredient	taken from
window + negative clip + area weighting; normalization to unit mass	Bodis *et al.* 2009 (bins)
physical blur width as a shift-tolerance ($\sigma = \sqrt{\ell^2 + d^2}$)	Castillo *et al.* 2013 (tree shift-insensitivity)
a peak/lineshape picture of the spectrum	Pierens *et al.* 2012 (NN peaks)
mean-centred zero-lag correlation (the discriminating step)	this work

References

1. L. Bodis, A. Ross, J. Bodis, E. Pretsch, *Automatic compatibility tests of HSQC NMR spectra with proposed structures of chemical compounds*, Talanta 79 (2009) 1379–1386. doi:10.1016/j.talanta.2009.06.017.
2. A.M. Castillo, L. Uribe, L. Patiny, J. Wist, *Fast and shift-insensitive similarity comparisons of NMR using a tree-representation of spectra*, Chemometrics and Intelligent Laboratory Systems 127 (2013) 1–6. doi:10.1016/j.chemolab.2013.05.009.
3. G.K. Pierens, S. Brossi, Z. Yang, D.C. Reutens, V. Vegh, *HSQC spectral based similarity matching of compounds using nearest neighbours and a fast discrete genetic algorithm*, Journal of Cheminformatics 4 (2012) 25. doi:10.1186/1758-2946-4-25.